

Fig. 9: 3D laser imaging concept

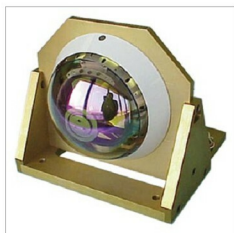


Fig. 10: Multimode-enhanced Ladar seeker

Fig. 9 illustrates the concept of flash-type 3D laser imaging. In this case, a diverging pulsed laser beam illuminates the entire scene of interest. The transmitted pulse time is referenced by an auxiliary photosensor. The back-scattered light is imaged onto a two-dimensional array of photosensing elements called pixels. While a conventional camera would measure the light intensity of back-scattered light pulse, in this case different sensor elements in the array measure time-of-flight. The time-of-flight is proportional to the distance between the point on the target from where the laser beam is reflected and the sensor element. The sensor array thus produces a 3D image of the target (angle-angle-range). An alternative approach of generating the target image is to use a scanning laser beam and a single sensor.

A common application of Ladar concept is in Ladar seekers, which are used mainly in conjunction with

other guidance systems for intended target discrimination from advanced decoys and aim-point selection. It is also well suited for combat identification, autonomous vehicle navigation and topography. Ladars are also suitable in finding targets hidden by camouflage nets and foliage.

Ladar seekers can detect and identify specific features of the target with very high definition, giving up to centimetre-level resolution from a few kilometres distance. An automatic target acquisition algorithm processes images to identify and acquire targets based on 3D templates stored in the weapon's memory before the mission. Ladar sensors are usually employed on loitering systems that look at the target from different angles, verify target's identity and select the best attack position for desired results.

Fig. 10 shows an advanced multimode Ladar seeker from Lockheed Martin. The seeker can operate in standalone semi-active laser (SAL) and Ladar modes as well as simultaneous SAL and Ladar modes for target identification, acquisition and tracking. It is designed to conduct wide-area search and identify actual or potential targets including those obscured by camouflage or foliage.

Such Ladar seekers have been successfully tested on Loitering Attack Missile (LAM) missions under DARPA's NLOS-LS (non-line-of-sight launch system) and USAF's LOCAAS (Low-cost Autonomous Attack System) programmes. LAM with its multimode Ladar seeker searches a large area and relays location of various targets back to the command centre, where these targets are engaged by direct attack or other assets. In case of a priority target, LAM can be commanded to break off its search mission and attack the target.

To be continued



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